

Quantification of the Pace of Biological Aging Among Participants in the US Health and Retirement Study

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I. Overview and Student Role

Chronological age is an imperfect measure of the aging process, which is shaped by genetic, environmental, and lifestyle factors. Biological age offers a more direct reflection of physiological state, helping identify individuals at elevated risk for age-related diseases and functional decline. Multiple approaches exist to estimate biological aging, from epigenetic clocks to composite biomarker scores, and new methods continue to emerge. We developed a new biological aging measure using the Health and Retirement Study cohort and a hierarchical model to capture shared slopes of biomarker change across multiple physiological systems. Across multiple aging-related outcomes, our slope-based measures generally outperformed or matched an existing biological aging measure developed in the same cohort, with a similar pattern for the DNA methylation–based constructs. Further analyses and external validation are planned for a future manuscript.

I led the programming of all analyses under the direct supervision of Marella, analyst within mentor-Belsky’s Geroscience Computational Core, and the overall direction of Belsky. I met through periodic meetings with Marella or Marella and Belsky to review progress and next steps.

II. Background

The Geroscience Hypothesis proposes that cellular-level changes underlying biological aging accumulate from early life, driving progressive functional decline and loss of resilience across systems, ultimately leading to chronic disease. Accelerated biological aging is therefore a potent risk factor for many chronic diseases and quantification of biological aging a powerful path to prevention. A critical public health problem is how to measure differences in the pace of biological aging before people become sick, in time for preventive intervention.

This study develops a new approach to estimate the pace of aging using longitudinal biomarker data from the U.S. Health and Retirement Study (HRS). The standard approach to measuring pace of aging is to independently model changes in each of a series of biomarkers representing different systems in the body and then combining estimates into a single index in a separate step. In contrast, our approach models changes in all biomarkers simultaneously to estimate their shared rate of change in a single step. The resulting measure better reflects the Geroscience Hypothesis construct of aging as a shared, underlying cause of changes across systems.

III. Methods

The Health and Retirement Study (HRS) is a nationally-representative longitudinal survey of U.S. residents aged 50 years and older and their spouses fielded every two years since 1992 (cumulative N=42,515). A new cohort of 51-56-year-olds and their spouses is enrolled every six years to maintain representativeness. Response rates across all waves range from 81-91%. We linked harmonized longitudinal HRS data curated by the RAND Corporation with biomarker data collected during home visits from 2006–2016 on two schedules to model pace of biological aging (**Figure 1**). To evaluate criterion validity, we tested associations with four outcomes: mortality, incident chronic disease, disability, and cognitive impairment. Mortality was assessed prospectively from baseline biomarker measurement through 2020. Incident chronic disease and disability (ADL/IADL limitations) were measured by self-report. Cognitive impairment was assessed using a standardized protocol. Incidence outcomes were defined through the 2020 wave or the participant's most recent follow-up. For epigenetic analysis, we used Health and Retirement Study DNA methylation data previously processed by Belsky Lab. These data included epigenetic clocks, estimated abundances of 12 white blood cell types, and site-level data at approximately one million genomic positions. Additional technical details are provided in **Supplemental Section 1**.

The more traditional 'Pace of Aging' measure from Balachandran et al., developed using the Health and Retirement Study (HRS), is derived from blood, physical, and functional biomarkers collected during home visits; we used the same curated dataset for our analyses. The dataset included three blood biomarkers (HbA1c, CRP and cystatin-C), three physical assessments (diastolic blood pressure, peak-flow lung-function testing and waist circumference), and three functional tests (gait speed, balance and grip strength). Following the prior analysis, we included waist circumference data from participants younger than age 65 years at baseline because of known nonlinear change at older ages. Participants received home visits at 4-year intervals (2006, 2010 and 2014 or 2008, 2012 and 2016). After applying inclusion criteria requiring at least two repeated measures of at least six of the nine biomarkers across all three data types and age below 90 years at the first biomarker measurement, our sample comprised 13,358 participants, of whom 41% were male.

We standardized biomarkers (**Figure 2**) to a common distribution by centering values on the sex-specific mean for participants aged <65 years (the mean age of participants in our sample) and dividing the centered values by the sex-specific standard deviation (SD) of that group (gait speed was measured in participants aged 65 years and older; we standardized values based on distributions for participants aged 65–75 years). Age-group-specific means and SDs for men and women included in analysis are reported in **Supplemental Section 2**. For biomarkers that decline with aging (diastolic blood pressure, peak flow, balance and grip strength), standardized values were reversed so that higher values corresponded to poorer organ-system function. Gait speed was

measured as the natural log of time to complete the walk test and did not need to be reversed.

We organized the dataset with one row per participant–biomarker–timepoint combination, containing standardized biomarker values, and applied a hierarchical linear model (HLM) to estimate trajectories of change. The model has a three-level structure: repeated measurements over time (Level 1) nested within biomarkers (Level 2), nested within participants (Level 3). Random intercepts and slopes at both higher levels allow individuals and biomarkers to vary in baseline levels and trajectories over time. The fixed intercept and slope represent the population-average baseline and rate of change. Parameters were estimated using restricted maximum likelihood (REML), providing unbiased estimates of variance components. The model can be written as:

$$y_{ibt} = (\beta_0 + \theta_{0i} + \delta_{0ib}) + (\beta_1 + \theta_{1i} + \delta_{1ib}) \cdot t + \varepsilon_{ibt} = \text{Intercept}_{ib} + \text{Slope}_{ib} \cdot t + \varepsilon_{ibt},$$

$$\text{Intercept}_{ib} = \beta_0 + \theta_{0i} + \delta_{0ib}, \quad \text{Slope}_{ib} = \beta_1 + \theta_{1i} + \delta_{1ib},$$

where y_{ibt} is the value of biomarker b for participant i at time t ; β_0 and β_1 are the population-average intercept and slope; θ_{0i} and θ_{1i} are participant-specific random intercepts and slopes; δ_{0ib} and δ_{1ib} are biomarker-specific random intercepts and slopes nested within participants; and ε_{ibt} is the residual error.

We constructed ‘Theta_Ratio’ (θ_{Ratio}) by summing fixed (β_1) and participant-specific random slopes (θ_{1i}) to obtain annual change shared across biomarkers and normalizing by the sex-specific mean among participants aged ≤ 65 , yielding a pace-of-aging measure relative to a reference population (1 = typical, > 1 faster, < 1 slower). We then derived ‘Theta_Ratio_GN’ ($\theta_{\text{Ratio,GN}}$) (**Figure 3** for its distribution) by calibrating θ_{Ratio} to mortality risk using sex-specific Gompertz survival models and scaling slopes by the ratio of slope- to age-associated hazards, producing our main measure: a mortality-normalized pace-of-aging metric. **Table 1** summarizes the aging measures.

Table 1: Glossary of Aging Measures in this Report

Measure	Description
PoA	Pace of Aging measure from Balachandran et al.—the relative biomarker-based pace vs. < 65 cohort (sex-specific); 1.1 = 10% faster.
Theta (θ_1)	The raw slope deviation from cohort mean from our model; cannot interpret as percent faster.
Theta_Ratio (θ_{Ratio})	Ratio of individual slope to cohort < 65 sex-specific mean; 1.1 = 10% faster.
Theta_Ratio_GN ($\theta_{\text{Ratio,GN}}$)	Mortality-calibrated pace built on Theta_Ratio; 1.1 = 10% faster in mortality-equivalent years.

To assess the contribution of each biomarker to the shared-change slope (θ_{1i}), we conducted a leave-one-out (LOO) analysis, removing one biomarker at a time, refitting the hierarchical model, and extracting participant-specific slopes. The slope for participant i in the model with biomarker

b left out is denoted $\theta_{1i}^{(-b)}$, which was then correlated with the full-model slope θ_{1i} .

We conducted mortality analysis by grouping people into tertiles of $\theta_{\text{Ratio,GN}}$ from slowest (lowest tertile) to fastest (highest tertile). Kaplan–Meier survival curves with 95% confidence intervals (CIs) were estimated for each tertile to examine differences in survival probability over time, with survival defined as follow-up until death or censoring. Pairwise Pearson correlations between $\theta_{\text{Ratio,GN}}$, PoA, and baseline age were computed using complete-case observations. Scatterplots with LOESS (locally estimated scatterplot smoothing) curves were used to visualize these relationships.

To build epigenetic “speedometers,” we used DNA methylation (DNAm) data. DNAm levels at thousands of CpG (cytosine–phosphate–guanine) sites, together with age and sex, were used to predict each outcome via ridge regression, which shrinks the influence of less informative sites while retaining all predictors in the model. The model was repeated 100 times on random 90% subsamples, and the resulting coefficients were averaged to generate stable DNAm-based scores for each individual. These scores summarize the aggregate methylation signal associated with each construct, analogous to established epigenetic clocks such as DunedinPACE (Belsky et al., 2022), and indicate which CpG sites consistently contributed to prediction. We applied this procedure to our theta-based constructs and PoA to obtain DNAm-based versions.

We examined associations of various biological aging measures with mortality, disease and disability, and cognitive decline. All clocks were residualized for baseline age, and models adjusted for baseline age and sex to capture variation due to biological aging beyond baseline age differences. To evaluate whether Gompertz-based scaling and normalization affects effect sizes, we included θ_1 , θ_{ratio} and $\theta_{\text{ratio,GN}}$ in the analyses along with PoA. Mortality risk was assessed using Cox proportional hazards models, reporting hazard ratios (HRs) with 95% CIs. Disease and disability changes over time were modeled via conditional Poisson regression, reporting rate ratios (RRs) with 95% CIs, and transitions between cognitive states were analyzed using multinomial logistic regression, reporting odds ratios (ORs) for worsening cognition with 95% CIs. For each clock, we computed the mean, standard deviation, range, correlation with age, and sex differences (standardized difference with 95% CI). Error-bar plots were used to visualize effect estimates. We repeated these analyses to compare the performance of PoA and $\theta_{\text{ratio,GN}}$ clocks with their DNAm counterparts. For each age-residualized clock–outcome pair, the same regression models with identical covariate adjustment were fit. Estimates were summarized in coefficient-style plots displaying point estimates and 95% CIs, with original (restricted to the 2,935 common IDs) and DNAm versions shown in adjacent rows within each outcome panel.

IV. Results

The hierarchical model included 280,693 observations from 13,358 participants and 115,061 biomarker-within-participant groups. Across the cohort, the average baseline biomarker value was 0.16 units, and the average annual change was 0.17 units ($p < 0.001$ for both). Random effects revealed substantial heterogeneity: participant-level intercepts and slopes varied with SDs of 0.37 and 0.16, respectively, and biomarker-level intercepts and slopes within participants varied with SDs of 0.75 and 0.29. The residual SD was 0.67. The leave-one-out analysis revealed high correlations (all near 1): the biomarker with the most influence (balance test, standardized) had $r = 0.92$ between full-model slopes and the model without it as seen in **Supplemental Section 3**.

The Kaplan–Meier curves (**Figure 4**) revealed that the fastest tertile of $\theta_{\text{Ratio,GN}}$ shows the fastest decline in survival probability. The survival probability for the fastest tertile starts falling sharply after four years of follow-up, and by the last follow-up time point, the difference between the fastest and middle tertiles is much larger than the difference between the middle and slowest tertiles. $\theta_{\text{Ratio,GN}}$ and PoA are strongly positively correlated, with similar associations for both males and females: approximately 81% (**Figure 5**). $\theta_{\text{Ratio,GN}}$ and baseline age are moderately correlated, with the strength of association differing by sex: 44% for men and 36% for women (**Figure 6**).

DNAm predictors explained substantial variance in PoA ($\text{CV } R^2 \approx 0.50$) and moderate variance in theta-based constructs ($\text{CV } R^2 \approx 0.22$) consistent with methylation capturing a partial but meaningful biological component of these phenotypes. All predictors relied on distributed CpG contributions typical of ridge regression, and age effects were lower for theta constructs compared to PoA. Further information regarding training results can be found in **Supplemental Section 4**.

Across most aging outcomes, the θ -based clocks (θ_1 , θ_{Ratio} , and $\theta_{\text{Ratio,GN}}$) demonstrated consistently slightly larger effect sizes than PoA (**Figure 7**). For mortality, hazard ratios were up to 1.55 for the θ measures versus 1.53 for PoA. Similar patterns were observed for cognitive impairment and functional decline: for CIND, effect sizes were ~ 1.34 versus 1.31, and for functional limitations, the θ clocks showed stronger associations for both ADLs (1.33 vs. 1.27) and IADLs (1.29 vs. 1.25). Effect sizes were nearly identical across θ , θ_{Ratio} , and $\theta_{\text{Ratio,GN}}$, indicating that Gompertz-based scaling and normalization did not alter predictive performance. θ clocks showed slightly weaker association only for dementia (1.42 vs. 1.46), whereas associations with chronic disease were weaker for all clocks, with PoA not reaching statistical significance.

We then examined observed and age-residualized DNAm-based constructs. Across outcomes and evaluated on the common participant IDs, DNAm-based measures generally matched or slightly exceeded the effect sizes of the original constructs. For PoA, comparing DNAm with the original version, effect sizes were: ADLs, 1.35 vs 1.21; IADLs, 1.30 vs 1.22; CIND, 1.30 vs 1.22; dementia,

1.77 vs 1.60; chronic disease, 1.04 vs 1.03; and death, 1.78 vs 1.59. For $\theta_{\text{Ratio,GN}}$, comparing DNAm with the original version, effect sizes were: ADLs, 1.38 vs 1.29; IADLs, 1.34 vs 1.32; CIND, 1.33 vs 1.28; dementia, 1.67 vs 1.52; chronic disease, 1.04 vs 1.03; and death, 1.79 vs 1.63 (**Figure 8**). Overall, DNAm $\theta_{\text{Ratio,GN}}$ outperformed DNAm PoA at predicting aging-related outcomes.

V. Discussion

We developed a new biological aging measure by extracting participant-specific biomarker slopes from our hierarchical model applied to the Health and Retirement Study, including 13,358 participants. Leave-one-out analyses indicated that none of the nine aging biomarkers dominated the construct, suggesting that the measure reflects a shared signal of multi-system biological aging. The slopes were ratio-transformed and Gompertz-normalized to place them on a common, interpretable scale. This mortality-calibrated measure was moderately correlated with participants' baseline age and strongly positively correlated with the Pace of Aging, indicating that higher values reflect faster biological aging relative to age at baseline. Survival analysis showed that participants in the fastest tertile experienced the most rapid decline in survival probability, whereas those in slower tertiles exhibited progressively higher survival, consistent with the measure's ability to stratify mortality risk. Overall, these findings demonstrate that our mortality-calibrated measure is a valid and reliable indicator of biological aging.

Except for dementia, our θ -based clocks generally slightly outperformed the established Pace of Aging measure (Balachandran et al.). We also mapped our mortality-calibrated construct onto DNA methylation (DNAm), alongside PoA, to generate their DNAm versions, and observed that the DNAm-based measures generally outperformed the original versions, with DNAm $\theta_{\text{Ratio,GN}}$ generally outperforming DNAm PoA across aging outcomes. The lower correlation with baseline age and the better or comparable performance of $\theta_{\text{Ratio,GN}}$ across multiple outcomes indicate that our mortality-calibrated θ measure captures biological aging more effectively. DNAm training further highlighted the stronger dependence of PoA on baseline age. Despite only moderate cross-validated R^2 , the DNAm-based $\theta_{\text{Ratio,GN}}$ measure retained strong associations with outcomes, suggesting that methylation signals isolate the risk-relevant component of the phenotype and that the machine learning procedure effectively reduced noise in the original measure.

Our new mortality-calibrated DNA methylation-based θ construct (DNAm $\theta_{\text{Ratio,GN}}$) can potentially function as an epigenetic clock with several applications: (1) identifying aging-related risk earlier in the life course; (2) complementing clinical, functional, and cognitive assessments by capturing biological information beyond baseline age; and (3) supporting intervention studies, longitudinal analyses, and personalized aging profiles. Limitations include the use of blood-derived methylation, which may not fully reflect tissue-specific aging, and unavoidable measurement error stemming

from cohort characteristics, survey design, and laboratory variability. Analyses adjusted only for age and sex; incorporating additional sociodemographic and lifestyle factors (e.g., smoking, race/ethnicity, education) may better account for confounding and heterogeneity. Furthermore, our nine-biomarker panel may not capture all aspects of biological aging. Finally, the clocks were developed and tested within the same cohort, which may limit generalizability.

Future work should include comparisons against established clocks within this cohort, as well as external validation in independent cohorts to assess generalizability. In particular, we plan to project the DNAm-based $\theta_{\text{Ratio,GN}}$ measure onto independent datasets (e.g., Framingham Heart Study, Women’s Health Initiative, CARDIA, CALERIE Trial) to test (a) associations with healthspan-related outcomes and (b) modifiability by interventions.

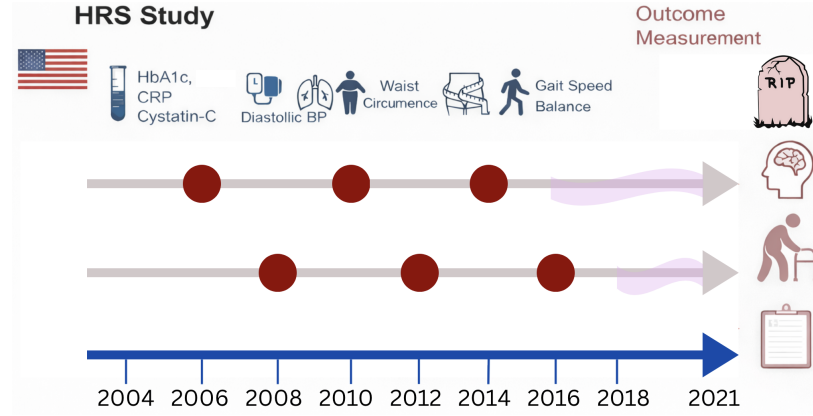


Figure 1: Study design for our analysis in the US Health and Retirement Study. Lines represent the two measurement schedules, each covering roughly half of the cohort. Circles indicate biomarker collection points, and pink ribbons show outcome data collection.

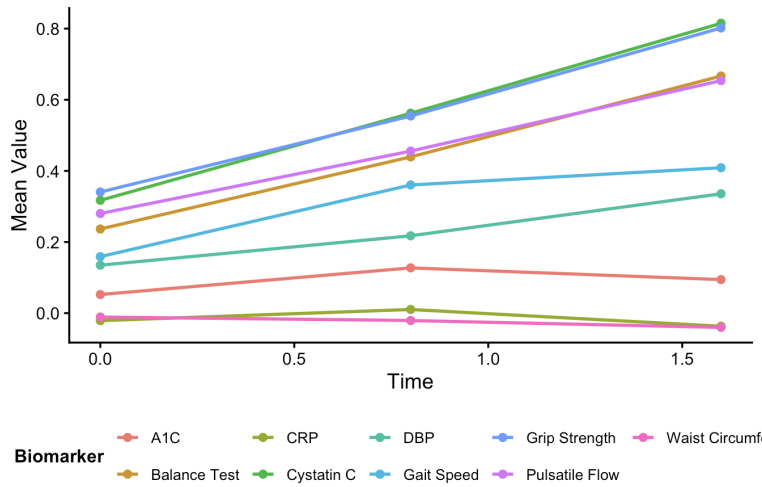


Figure 2: Trajectories of the nine standardized aging biomarkers in the Health and Retirement Study (HRS) cohort used to derive the Pace of Aging (Balachandran et al.) and Theta-based clocks.

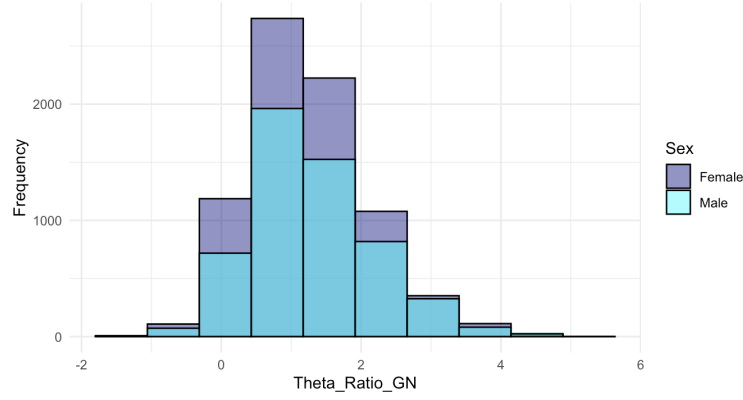


Figure 3: Sex-specific distribution of *Theta_Ratio_GN*, with the female histogram in the background and male values overlaid.

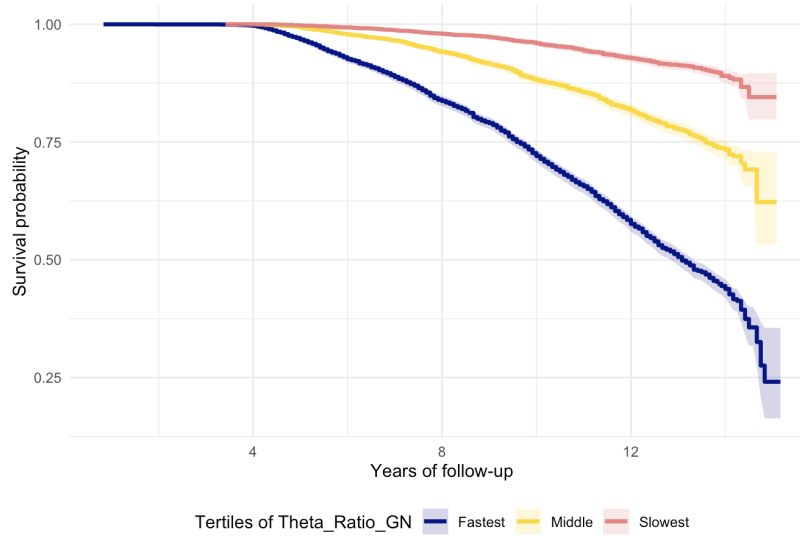


Figure 4: Kaplan-Meier curve for tertiles of *Theta_Ratio_GN*, showing survival probability over follow-up time to death or censoring. Shaded areas represent 95 % confidence intervals.

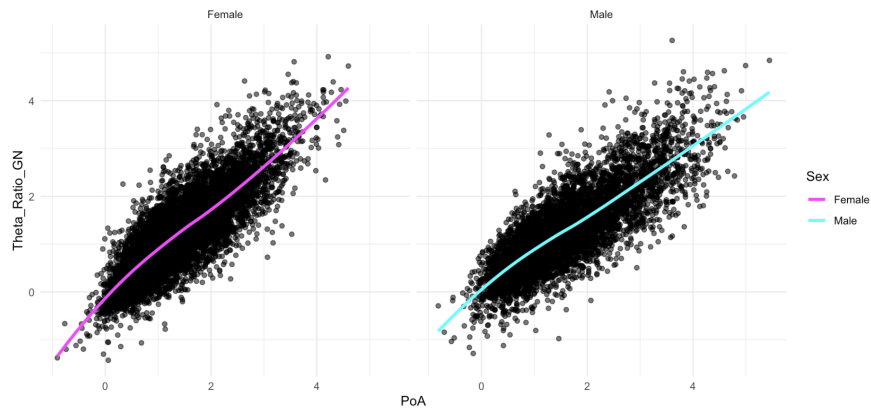


Figure 5: Correlation plot between *Theta_Ratio_GN* and Pace of Aging (Balachandran et al.) across sexes. Points represent individual participants, and the solid line shows the LOESS-smoothed fit.

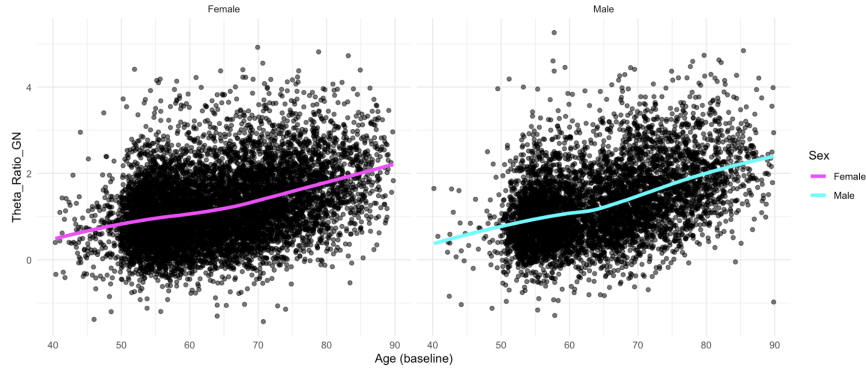


Figure 6: Correlation plot between *Theta_Ratio_GN* and baseline age across sexes. Points represent individual participants, and the solid line shows the LOESS-smoothed fit.

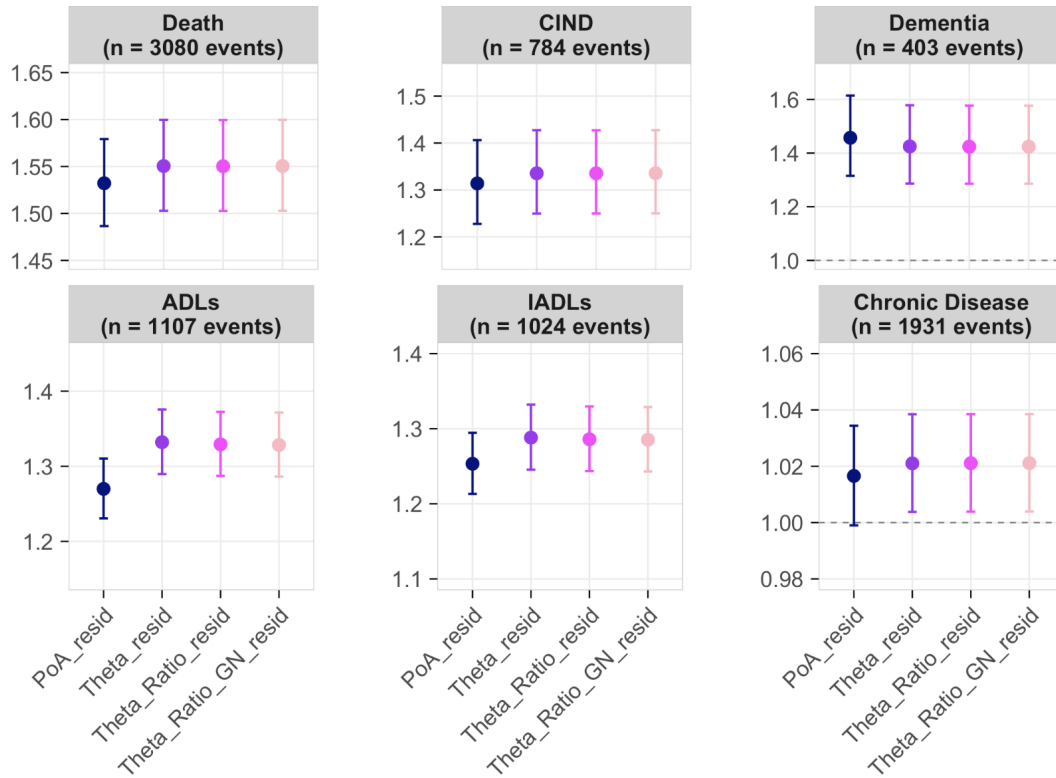


Figure 7: Effect sizes with 95% confidence intervals for age-residualized Pace of Aging (Balachandran et al.) and Theta-based clocks across multiple aging outcomes, showing the strength of association for each measure.

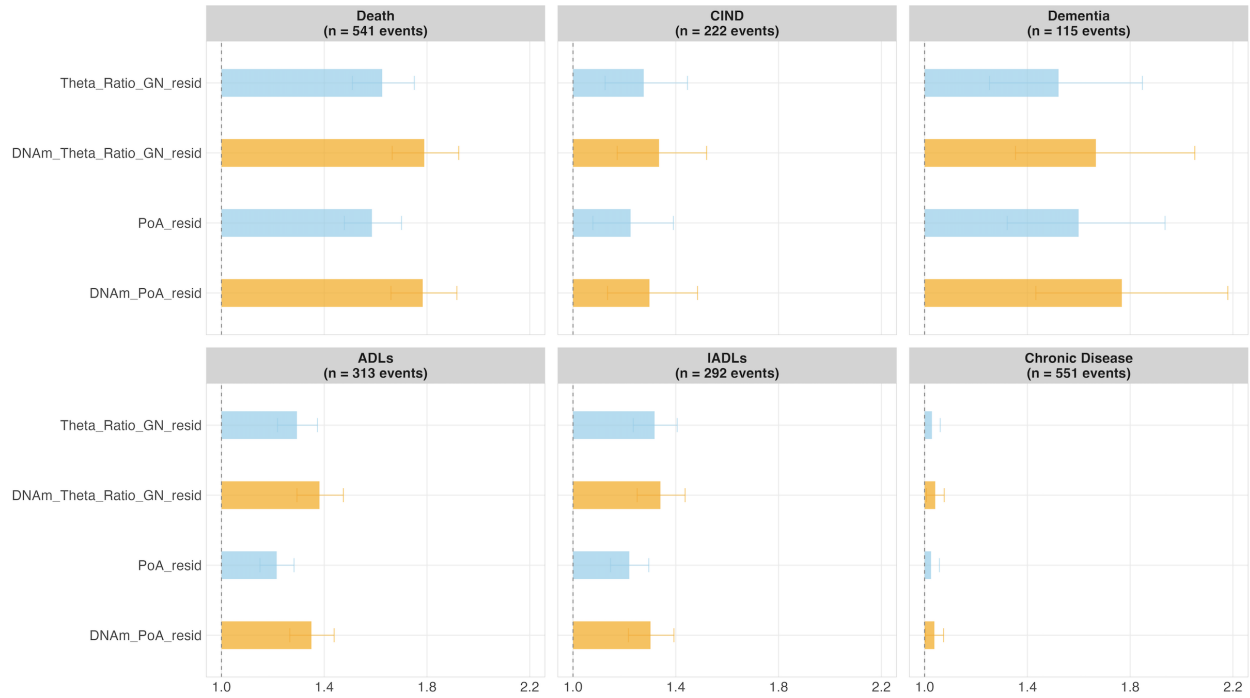


Figure 8: Comparison of effect sizes for age-residualized Pace of Aging (Balachandran et al.) and Theta_Ratio_GN with their DNA methylation (DNAm) versions across multiple aging outcomes.

References

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Supplementary Material

Supplemental Section 1: DNA Methylation Data Processing

Raw IDAT files were obtained from the HRS secure download portal and processed in an AWS virtual machine through the Ronin interface. IDATs were matched to phenotype information and partitioned into five chunks of roughly equal size. When present, replicate samples were kept together in the same chunk. Preprocessing and normalization of DNA methylation data were conducted in R (v 4.4.1) on $n = 4018$ samples using minfi (v1.5.0) and ewastools packages (v 1.7.2). Sample-level quality control was performed using 23 binary pass/fail indicators spanning a range of methylation array QC metrics. Seventeen of these metrics are derived from Illumina GenomeStudio control probes, as implemented in the ewastools package following standard protocols. An additional six QC measures were included based on implementations in minfi and waterRmelon: (1) signal quality (samples with methylated or unmethylated intensity < 10.5 flagged as failed), (2) bead count (samples with $> 5\%$ of probes having < 5 beads), (3) bisulfite conversion efficiency ($< 80\%$), (4) detection p -value (average p -value > 0.05), (5) sex mismatch (discrepancy between predicted and reported sex based on X and Y methylation), and (6) SNP outlier detection (based on clustering of SNP probe intensities, with scores ≥ 4 indicating likely contamination or degradation). $n = 535$ samples failed sample QC. Normalization was performed using the preprocessNoob function in the minfi package (v1.50.0). Epigenetic clocks were calculated on Noob-normalized data for original clocks and principal-components based versions, after sample QC but prior to probe QC. DunedinPACE was derived using the DunedinPACE package. Cell composition was estimated via a pipeline customized for EPICv2 arrays, using the minfi estimateCellCounts2 function and the Salas extended reference panel. Technical principal components were based on principal components analysis of control probes from samples that passed QC. Probe QC was carried out on 865,859 probes for each chunk separately. Probes were removed if they contained SNPs at the CpG site or single-base extension, were flagged as cross-hybridizing or polymorphic, mapped to X or Y chromosomes, had beadcounts < 3 in $> 5\%$ of samples, or showed an average detection p -value ≥ 0.05 in $> 1\%$ of samples. After QC, 776,516 probes remained for analysis. Finally, intra-sample normalization using beta-mixture quantile (BMIQ) was carried out.

Supplemental Section 2: Biomarker Summary Statistics

N refers to the total number of observations. SD refers to the standard deviation.

eTable 1: Overall Biomarker Summary Statistics (Baseline, N=13,358)

Biomarker	N	Mean	SD
HbA1c	12,769	5.77	0.85
log C-reactive Protein	13,033	0.69	1.17
log Cystatin-C	12,899	0.69	0.15
Diastolic blood pressure	13,125	80.00	11.50
Waist circumference	13,113	39.64	6.05
Peak Flow	13,240	378.88	130.60
Grip Strength	13,041	33.56	11.18
Balance Test	13,006	2.71	0.53
log Gait Speed	6,449	1.40	0.25

eTable 2: Female Biomarkers by Age Group

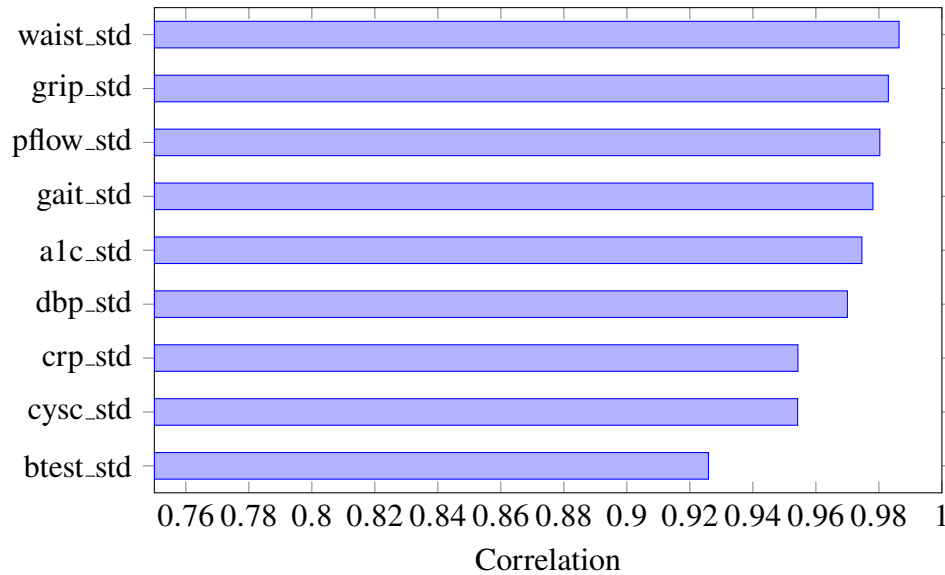
Biomarker	<65 N	<65 Mean (SD)	≥65 N	≥65 Mean (SD)
HbA1c	4,044	5.73 (0.85)	3,444	5.81 (0.77)
log C-reactive Protein	4,226	0.88 (1.20)	3,444	0.73 (1.16)
log Cystatin-C	4,177	0.65 (0.13)	3,418	0.74 (0.16)
Diastolic blood pressure	4,239	80.72 (11.61)	3,410	78.05 (11.33)
Waist circumference	4,247	38.80 (6.59)	3,386	38.17 (5.91)
Peak Flow	4,309	343.11 (85.27)	3,438	284.95 (80.64)
Grip Strength	4,196	28.67 (6.07)	3,388	24.11 (5.54)
Balance Test	4,262	2.77 (0.47)	3,341	2.52 (0.64)
log Gait Speed	-	-	3,758	1.43 (0.26)

eTable 3: Male Biomarkers by Age Group

Biomarker	<65 N	<65 Mean (SD)	≥65 N	≥65 Mean (SD)
HbA1c	2,747	5.73 (0.95)	2,534	5.84 (0.81)
log C-reactive Protein	2,832	0.49 (1.13)	2,531	0.56 (1.12)
log Cystatin-C	2,789	0.65 (0.12)	2,515	0.74 (0.15)
Diastolic blood pressure	2,912	82.71 (11.24)	2,564	78.33 (11.09)
Waist circumference	2,924	41.03 (5.70)	2,556	41.38 (4.80)
Peak Flow	2,922	497.69 (126.71)	2,571	429.39 (127.22)
Grip Strength	2,887	46.29 (8.89)	2,570	39.70 (8.24)
Balance Test	2,895	2.86 (0.40)	2,508	2.68 (0.55)
log Gait Speed	-	-	2,719	1.37 (0.23)

Supplemental Section 3: Leave One Biomarker Out Analysis Results

The slope for participant i in the model with biomarker b left out is denoted $\theta_{1i}^{(-b)}$, which was then correlated with the full-model slope θ_{1i} .

eFigure 1: Correlation Graph

Y-axis shows the biomarker excluded in each leave-one-out LOO HLM. X-axis shows the correlation between participant-specific slopes estimated in the LOO model, $\theta_{1i}^{(-b)}$, and the corresponding full-model slopes, θ_{1i} .

eTable 4: Correlation Table

Biomarker Left Out	Correlation
waist_std	0.9864
grip_std	0.9830
pflow_std	0.9803
gait_std	0.9781
alc_std	0.9746
dbp_std	0.9700
crp_std	0.9543
cysc_std	0.9542
btest_std	0.9259

Supplemental Section 4: DNAm Training Results**eTable 5: DNAm Training Results for PoA and Theta-based Clocks**

Training Results Metric	PoA	θ_1	θ_{Ratio}	$\theta_{\text{Ratio,GN}}$
Best Lambda	5.545	5.460	3.270	4.687
CV R^2	0.496	0.224	0.222	0.222
Final Ensemble Features	14514	14514	14514	14514
Final Ensemble R^2	0.763	0.714	0.716	0.716
Age Coefficient	0.051	0.016	0.010	0.014

Best Lambda – The value of the regularization parameter (λ) that gave the best model performance during training.

CV R^2 – Cross-validated coefficient of determination; measures how well the model predicts held-out data.

Final Ensemble Features – Number of features (biomarkers or derived variables) included in the final ensemble model.

Final Ensemble R^2 – Coefficient of determination for the final model on the training set; shows overall model fit.

Age Coefficient – The estimated effect of age on the outcome in the final model.